

PART 1 REPORT

Data Warehouse Design, ETL and OLAP Analysis

1. ETL Procedure

The raw CSV files were first loaded into staging tables in DuckDB. The data was then cleaned and transformed before loading it into the final Snowflake schema.

The geography hierarchy was created from the geolocation data. Since the region level was not present in the original data, the Brazilian states were mapped to five regions. The hierarchy was loaded from the top level down as:

$$Region \rightarrow State \rightarrow City \rightarrow ZIPPrefix$$

City names and ZIP prefixes were cleaned before loading them into the dimension tables. The same geography hierarchy was used for both customers and sellers.

For the product data, products were linked to their categories and the English category names were included in the category dimension.

Seller acquisition information was obtained by joining the leads and closed-deals data. The seller dimension was then linked to the acquisition information and the seller's ZIP prefix.

For the fact table, order items were combined with order, payment and review information. Payment information was aggregated at the order level. Reviews and delivery dates were joined using left joins so that orders with missing reviews or missing delivery dates were not removed.

The fact table was kept at the order-item level and revenue was calculated using:

$$Revenue = Price + FreightValue$$

The dimension tables were loaded before the fact table so that the required foreign-key relationships could be maintained.

2. Normalised Hierarchies

The following hierarchies were normalised into separate tables:

- **Geography:** ZIP Prefix $\rightarrow$ City $\rightarrow$ State $\rightarrow$ Region
- **Product:** Product $\rightarrow$ Category
- **Seller:** Seller $\rightarrow$ Acquisition Details

The geography hierarchy was separated because the same state, region and city information would otherwise be repeated for many customers and sellers. It also allows analysis at different geographical levels using roll-up and drill-down operations.

The product hierarchy was separated so that category information and its English translation are stored once and can be referenced by multiple products.

Seller acquisition details were separated because lead origin, business segment and won date describe the seller's acquisition rather than the seller itself.

3. OLAP Analysis and Insights

The analytical queries were used to study revenue across time, product categories, geography, customers, sellers, delivery performance and seller acquisition channels.

3.1 Revenue by Time

The total revenue obtained from the fact table was approximately 15.84 million.

Year	Total Revenue
2016	57,183.21
2017	7,142,672.43
2018	8,643,697.60

Revenue increased considerably from 2016 to 2017 and increased further in 2018. The highest-revenue month was October in 2016 with 56,808.84, November in 2017 with 1,179,143.77, and April in 2018 with 1,159,698.04.

3.2 Product Categories

The *health_beauty* category generated the highest revenue at 1,441,248.07. The next highest categories were *watches_gifts* with 1,305,541.61 and *bed_bath_table* with 1,241,681.72.

The category-to-product drill-down was used to move from category-level revenue to individual products and identify the products contributing to the category totals.

3.3 Geographical Analysis

The geography roll-up showed that the Southeast was the highest-revenue region, generating approximately 10.22 million. The South was the second-highest region with approximately 2.30 million, followed by the Northeast with approximately 1.87 million.

Within the Southeast, São Paulo had the highest state-level revenue at approximately 5.92 million. At the city level, São Paulo generated approximately 2.15 million.

The geography query covers the records for which the customer geography could be mapped successfully. Therefore, its total is slightly lower than the overall fact-table revenue.

3.4 Customers and Sellers

The top five customers were identified using their total purchase amount. The highest-spending customer in the result had a total purchase amount of 13,664.08.

The seller analysis compared sellers within their respective states. The highest-revenue seller in the results generated 249,640.70 in São Paulo. The next highest examples were a seller in Bahia with 235,856.68 and a seller in Rio de Janeiro with 139,909.69.

3.5 Region, Category and Year Analysis

The CUBE analysis was used to examine revenue at different combinations of region, category and year.

The highest detailed combination in the result was the Southeast region, *health_beauty* category and 2018, with revenue of 562,357.43. Other high-revenue combinations included Southeast and *bed_bath_table* in 2018 with 493,721.58 and Southeast and *watches_gifts* in 2018 with 484,144.12.

3.6 Delivery Performance and Reviews

The average review score was compared for early, on-time and late deliveries.

Delivery Status	Average Review Score	Orders
Early	4.294	88,168
On Time	4.034	1,280
Late	2.271	6,382

The results show that early deliveries received the highest average review score, while late deliveries received a considerably lower score. The average score for late deliveries was 2.271 compared with 4.034 for on-time deliveries. This indicates a clear relationship between late delivery and lower customer review scores.

3.7 Seller Acquisition Channels

Among the lead origins in the query output, *organic_search* generated the highest total seller revenue at 235,956.58 from 113 sellers. It was followed by *paid_search* with 182,000.06 from 101 sellers and *social* with 51,363.47 from 31 sellers.

Therefore, organic search was the strongest revenue-generating seller acquisition channel among the channels present in the result.

4. Conclusion

The Snowflake schema separates the geography, product and seller acquisition hierarchies while keeping the fact table at the order-item level. The ETL process cleaned the raw data, established the required relationships and retained records with missing delivery dates or reviews.

The OLAP analysis showed that revenue increased substantially after 2016, the Southeast was the strongest region, and health and beauty was the highest-revenue product category. The analysis also showed that late deliveries were associated with much lower review scores. Among the seller acquisition channels, organic search generated the highest revenue.